HENKEL CORPORATION

EMERY GROUP

Material Safety Data Sheet

EMERGENCY PHONE: (513) 482-2297 **CHEMTREC** 800-424-9300

MSDS REFERENCE: EMERY 3004 (7/10/95)

SECTION I - IDENTIFICATION

PRODUCT: EMERY 3004 SYNTHETIC HYDROCARBON 4 CST FLUID

SYNONYMS: POLYALPHAOLEFIN

CHEMICAL: SYNTHETIC ALIPHATIC HYDROCARBON

CAS NO: 68649-12-7

SARA HAZARD: NONE NOTED (SECTION 311/312)
TITLE III SECTION 313- NOT LISTED

SECTION II - INGREDIENTS AND HAZARD CLASSIFICATION

COMPOSITION	%	PEL/TLV	HAZARD
POLYALPHAOLEFIN (68649-12-7)	100	NONE/NONE	NONE NOTED

SECTION III - HEALTH INFORMATION

INHALATION: THE ESTIMATED LC50 FOR A 1 HOUR EXPOSURE TO 2 CST PAO WAS 4.68 MG/L (RATS), WHICH IS CONSIDERED TOXIC. IN ORDER TO DETERMINE THE LC50 VALUE, EXTREMELY HEAVY MISTS OF PAO WERE REQUIRED. THE VERY HEAVY MISTS AT THE NECESSARY CONCENTRATION MADE VISIBILITY DIFFICULT AND WOULD BE DIFFICULT TO WORK IN FOR ANY PERIOD OF TIME. THE AUTHOR REPORTED THAT HISTOPATHOLOGICAL CHANGES MAY HAVE BEEN A RESPONSE TO A PHYSICAL INSULT RATHER THAN A SPECIFIC COMPOUND RELATED TOXICITY EFFECT AND THAT 2 CST PAO MAY BE CONSIDERED NON-HAZARDOUS FOR ALL PRACTICAL PURPOSES BY INHALATION.

INGESTION: THE ACUTE ORAL LD50 VALUE WAS FOUND TO BE GREATER THAN 5.0 G/KG IN MALE AND FEMALE SPRAGUE-DAWLEY RATS. THE MATERIAL IS NOT CLASSIFIED AS TOXIC BY ORAL ADMINISTRATION AS DEFINED IN 16 CFR 1500.

EYE CONTACT: THE EYES OF 2 RABBITS WERE FOUND TO SHOW EVIDENCE OF CONJUNCTIVAL CHANGES. IRRITATION SCORES IN INDIVIDUAL RABBITS RANGED FROM 0-2 (SCALE 0-110). THE MATERIAL IS NOT CLASSIFIED AS AN IRRITANT BY OCULAR APPLICATION AS DEFINED BY 16 CFR 1500.

SKIN CONTACT: THE PRIMARY IRRITATION INDEX WAS FOUND TO BE 1.3 (SCALE 0-8) BASED ON ERYTHEMA AND EDEMA. NO EVIDENCE OF TISSUE DAMAGE WAS FOUND. THE MATERIAL IS NOT CLASSIFIED AS A PRIMARY IRRITANT OR AS A CORROSIVE BY DERMAL APPLICATION AS DEFINED BY 16 CFR 1500.

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TESTING OF EMERY 3004 FOR COMEDOGENIC POTENTIAL (BLACKHEAD FORMATION) INDICATED THAT UNDER THE CONDITIONS OF THE STUDY, THE MATERIAL WAS NOT CONSIDERED TO SHOW COMEDOGENIC POTENTIAL.

SECTION IV - OCCUPATIONAL EXPOSURE LIMITS

PEL: TWA: 5 MG/M3 (OIL MIST)

TLV: TWA: 5 MG/M3; STEL: 10 MG/M3 (OIL MIST)

SECTION V - EMERGENCY FIRST AID PROCEDURE

FOR OVEREXPOSURE BY SWALLOWING: DO NOT INDUCE VOMITING. IF VICTIM IS CONSCIOUS AND ABLE TO SWALLOW, PROMPTLY HAVE VICTIM DRINK WATER TO DILUTE. DO NOT GIVE SODIUM BICARBONATE, FRUIT JUICES OR VINEGAR. NEVER GIVE ANYTHING BY MOUTH IF THE VICTIM IS UNCONSCIOUS OR HAVING CONVULSIONS. **CALL A PHYSICIAN OR POISON CONTROL CENTER IMMEDIATELY.**

FOR OVEREXPOSURE BY SKIN CONTACT: WASH AFFECTED AREA.

FOR OVEREXPOSURE BY EYE CONTACT: IMMEDIATELY FLUSH EYES WITH PLENTY OF COOL WATER FOR AT LEAST 15 MINUTES. DO NOT LET VICTIM RUB EYES.

FOR OVEREXPOSURE BY INHALATION: IMMEDIATELY REMOVE VICTIM TO FRESH AIR. IF VICTIM HAS STOPPED BREATHING GIVE ARTIFICIAL RESPIRATION, PREFERABLY BY MOUTH-TO-MOUTH. **GET MEDICAL ATTENTION IMMEDIATELY.**

SECTION VI - PHYSICAL DATA

BOILING POINT: 754 DEG F (401 DEG C) ASTM D-86
VAPOR PRESSURE: <1 MM HG PRESSURE AT 20 DEG C
SPECIFIC GRAVITY: 0.819 AT 15.6/15.6 DEG C
SOLUBILITY IN WATER: INSOLUBLE
APPEARANCE AND COLOR:
COLORLESS, ODORLESS FLUID

SECTION VII - FIRE AND EXPLOSION HAZARDS

FLASH POINT & METHOD USED: 432 DEG F (222 DEG C) ASTM D-92
FLAMMABLE LIMITS IN AIR, % BY VOL. LOWER: NOT DETERMINED
FLAMMABLE LIMITS IN AIR, % BY VOL. UPPER: NOT DETERMINED
NFPA RATING: NO NFPA RATING
HMIS RATING: HEALTH (1) FIRE (1) REACTIVITY (0)
SPECIAL FIRE FIGHTING PROCEDURES & PRECAUTIONS
(INDIVIDUALS SHOULD PERFORM ONLY THOSE FIRE FIGHTING PROCEDURES FOR WHICH THEY HAVE BEEN TRAINED). USE WATER SPRAY, DRY CHEMICAL, FOAM OR CARBON DIOXIDE. WATER MAY BE INEFFECTIVE BUT SHOULD BE USED TO KEEP FIRE-EXPOSED CONTAINERS COOL. IF A SPILL OR LEAK HAS NOT IGNITED, USE WATER SPRAY TO DISPERSE THE VAPORS. WATER SPRAY MAY BE USED TO FLUSH SPILLS AWAY FROM FIRE.
UNUSUAL FIRE & EXPLOSION HAZARDS
FIREFIGHTERS SHOULD WEAR SELF-CONTAINED BREATHING APPARATUS IN THE POSITIVE-PRESSURE MODE WITH A FULL FACEPIECE WHEN THERE IS A POSSIBILITY OF EXPOSURE TO SMOKE, FUMES OR HAZARDOUS DECOMPOSITION PRODUCTS.

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SECTION VIII - REACTIVITY

STABILITY:
GENERALLY STABLE
HAZARDOUS POLYMERIZATION:
NONE LIKELY
CONDITIONS & MATERIALS TO AVOID:
AVOID HEATING TO DECOMPOSITION. THE USER IS ADVISED TO HAVE A SAFETY EXPERT EVALUATE THE SPECIFIC CONDITIONS OF USE.
HAZARDOUS DECOMPOSITION PRODUCTS:
DECOMPOSITION MAY PRODUCE CARBON MONOXIDE AND CARBON DIOXIDE.

SECTION IX - EMPLOYEE PROTECTION

CONTROL MEASURES:
HANDLE IN THE PRESENCE OF ADEQUATE VENTILATION.
RESPIRATORY PROTECTION:
WHERE EXPOSURE IS LIKELY TO EXCEED ACCEPTABLE CRITERIA (SEE SECTIONS II AND IV), USE NIOSH/OSHA APPROVED RESPIRATORY EQUIPMENT. RESPIRATORS SHOULD BE SELECTED BASED ON THE FORM AND CONCENTRATION OF CONTAMINANT IN AIR AND IN ACCORDANCE WITH OSHA (29 CFR 1910.134).
PROTECTIVE CLOTHING:
WEAR GLOVES AND PROTECTIVE CLOTHING WHICH ARE IMPERVIOUS TO THE PRODUCT FOR THE DURATION OF ANTICIPATED EXPOSURE IF THERE IS POTENTIAL FOR PROLONGED OR REPEATED SKIN CONTACT.
EYE PROTECTION:
WEAR SAFETY GLASSES MEETING THE SPECIFICATIONS OF ANSI STANDARD Z87.1.

SECTION X - ENVIRONMENTAL PROTECTION

ENVIRONMENTAL PRECAUTIONS:
AVOID UNCONTROLLED RELEASES OF THIS MATERIAL. WHERE SPILLS ARE POSSIBLE, A COMPREHENSIVE SPILL RESPONSE PLAN SHOULD BE DEVELOPED AND IMPLEMENTED.
SPILL OR LEAK PRECAUTIONS:
WEAR APPROPRIATE RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING AS DESCRIBED IN SECTION IX. CONTAIN SPILLED MATERIAL. TRANSFER TO SECURE CONTAINERS. WHERE NECESSARY, COLLECT USING ABSORBENT MEDIA.
IN THE EVENT OF AN UNCONTROLLED RELEASE OF THIS MATERIAL, THE USER SHOULD DETERMINE IF THE RELEASE IS REPORTABLE UNDER APPLICABLE LAWS AND REGULATIONS.
WASTE DISPOSAL:
ALL RECOVERED MATERIAL SHOULD BE PACKAGED, LABELED, TRANSPORTED, AND DISPOSED OR RECLAIMED IN CONFORMANCE WITH APPLICABLE LAWS AND REGULATIONS AND IN CONFORMANCE WITH GOOD ENGINEERING PRACTICES. AVOID LANDFILLING OF LIQUIDS. RECLAIM WHERE POSSIBLE.

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SECTION XI - REGULATORY CONTROLS

DEPARTMENT OF TRANSPORTATION:
DOT CLASSIFICATION: NOT REGULATED
DOT PROPER SHIPPING NAME:
OTHER DOT INFORMATION:
OTHER REGULATORY REQUIREMENTS:
LISTED IN TSCA INVENTORY
CERCLA HAZARDOUS MATERIALS:
NONE NOTED

SECTION XII - PRECAUTIONS: HANDLING, STORAGE AND USAGE

NO SPECIAL PRECAUTIONS NECESSARY.

The information presented herein is believed to be factual as it has been derived from the works and opinions of persons believed to be qualified experts; however, nothing contained in this information is to be taken as a warranty or representation for which Henkel Corporation bears legal responsibility. The user should review any recommendations in the specific context of the intended use to determine whether they are appropriate.

PREPARED BY: ROBERT E. BORGERDING

DATE: 7/10/95
SUPERSEDES: 12/3/92

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CERTIFICATE OF ANALYSIS
ATI PAO-4

LOT # DCS-072315

PROPERTY	SPECIFICATION	TEST METHOD	RESULT
Specific Gravity @ 60F/60F	Report	ASTM D4052	0.8191
Viscosity @ 100 °C, cSt	3.80 to 4.00	ASTM D7042	3.82
Viscosity @ 40 °C, cSt	Report	ASTM D7042	16.71
Viscosity @-40 °C, cSt	<2800	ASTM D445	2485
Flash Point COC, °C	>210	ASTM D92	228
Pour Point, °C	<-60	ASTM D5950	-69
Water, ppm-WT	<50	ASTM E1064	31
Appearance -40, °C	Bright and Clear	ASTM D445	Bright and Clear
Color, Pt/Co	<25	ASTM D5386	4

Date of Manufacture: 7/22/15

Shelf Life: Ten (10) years from date of shipment when properly stored.

Store in a temperature controlled, dry, environment between 15 and 30 °C
Avoid exposure to UV or direct sunlight

ATI PAO-4 is 4cSt polyalphaolefin and does not contain animal parts or by-products

Approved by: _____ Date: 9/23/15
Quality Assurance Manager

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